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Q1. Overfitting - training accuracy high, test accuracy low; the model learnt the noise.
Methods: regularisation (penalise big weights), getting more training data.

Q2. $Q = XW_q$, $K = XW_k$, $V = XW_v$.
Attention = $\text{softmax}(QK^T / \sqrt{d_k})\, V$.
Softmax makes the weights between 0 and 1.

Q3. $P = 0.8$, $R = 0.667$, $F1 = 2(0.8)(0.667)/(1.467) = 0.727$, acc = 0.7

Q4. (a) GAN adversarial, diffusion denoising. (b) GANs unstable. (c) GAN mode collapse. (d) GAN faster.

Q5. Index the docs as embeddings; retrieve the similar chunks for a query; the LLM generates the answer from them.

Failure: hallucination - prompt it to only use the context.